

Review

Advances in copper pillar bumps for high-density electronic packaging

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ABSTRACT

With the deceleration of Moore's Law, the improvement of chip performance is transitioning from pure transistor scaling to the parallel advancement of device miniaturization and advanced packaging technologies, aiming to achieve higher levels of system integration and interconnect density. Following this trend, copper pillar bumps, owing to their miniature dimensions, fine pitch, and outstanding electrical and thermal properties, have emerged as essential interconnect structures in 2.5D and 3D packaging technologies. However, copper pillar bump technology continues to encounter issues, including elevated interconnection temperatures, inadequate thermo-electric reliability, and constrained performance under ultra-high-density conditions. This review provides an overview of the research background, current progress, and prospective trends in copper pillar bump technology. Beginning with the basic fabrication processes, this paper elaborates on the evolution of material systems for copper pillar bumps and the most recent advancements in novel fabrication techniques. Next, the underlying mechanisms and suitability of different interconnection approaches are examined. Moreover, advances in reliability testing, failure mechanisms, and optimization strategies for interconnections are summarized, providing theoretical guidance for extending their service life and enhancing long-term reliability. Finally, the prospects and research directions of copper pillar bump technology are discussed.

1. Introduction

In recent years, the rapid development of artificial intelligence (AI), high-performance computing (HPC), autonomous driving and intelligent vehicles, as well as 5G/6G communications, has placed increasing demands on semiconductor technologies, including higher computing power and bandwidth, lower power consumption and latency, and better thermal management. To meet these requirements, integrated circuits (ICs) are evolving toward higher integration and miniaturization [1–3]. In particular, in the field of high-performance chips, such as graphics processing units (GPUs) and central processing units (CPUs), continuous improvement in transistor density and optimization of packaging structures are required to satisfy computing power and thermal management demands [4,5]. However, as semiconductor manufacturing processes approach the limits of Moore's Law [6,7], merely scaling down the process nodes can no longer deliver significant improvements in chip performance. Against this backdrop, with the continuous advancement and co-evolution of interconnection technologies, semiconductor packaging structures have gradually transitioned from two-dimensional to three-dimensional configurations to achieve higher density and integration [8,9].

In the evolution of packaging technology, the initial form of interconnection is wire bonding, in which metal wires (Al, Au, or Cu) are used to connect the active surface of the chip to the substrate leads for electrical signal transmission. This method features a simple process but suffers from drawbacks such as long transmission paths and large package sizes [10]. To overcome these limitations, IBM proposed the Controlled Collapse Chip Connection (C4) technology in 1960, in which solder bumps are formed on the active surface of the chip and directly connected to the substrate using a flip-chip configuration. Unlike wire bonding, which routes connections along the chip edges, C4 bumps can be distributed across the entire chip surface, significantly increasing packaging density and reducing the electrical signal transmission path [11]. However, the collapse and non-uniform deformation of C4 bumps during reflow likely lead to bridging between adjacent bumps or the formation of voids during underfill processes. Moreover, since the bumps serve as both mechanical interconnects and electrical conduits, they face severe challenges in thermal reliability and electromigration under high-density packaging conditions [12,13].

To meet the packaging requirements for smaller size, higher performance, and improved reliability, the copper pillar bump (CPB) technology was introduced [14,15]. Compared with C4 bumps, this

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technology forms only a small amount of solder on the top of the copper pillar, resulting in less deformation during bonding. The bump primarily serves as a mechanical interconnect, while the copper pillar, owing to its excellent electrical and thermal conductivity [16] and strong resistance to electromigration [17], effectively mitigates the effects of thermal and electrical stresses on the solder joint [18]. In addition, the uniformity of copper pillar height and morphology contributes to a more consistent underfill process. These advantages make CPB technology a key solution for high-density, fine-pitch packaging applications (20–100 μm) [19, 20]. Fig. 1 illustrates a typical 2.5D packaging structure. In this structure, the high-bandwidth memory (HBM) module uses copper pillar micro-bumps to connect through-silicon vias (TSVs) between different chips. The interconnections between the chip and the interposer, as well as between the interposer and the substrate, can be achieved using copper pillar bumps of varying sizes and pitches according to density requirements. This demonstrates the critical role of copper pillar bumps in high-density interconnection structures at the application level.

As a key interconnection technology in advanced packaging, CPB technology has attracted significant research attention in recent years. Although several reviews have discussed CPB or related interconnection technologies [21–23], these studies mainly focus on specific aspects, such as comparisons with conventional C4 technology, fabrication processes, or reliability. A systematic understanding that integrates material systems, interconnection methods, and reliability performance is still lacking.

Accordingly, this work aims to provide a more comprehensive and integrated perspective on CPB technology (Fig. 2). In addition to conventional electroplating-reflow processes, emerging fabrication strategies based on micro-nano material systems are reviewed. Interconnection methods are discussed across different pitch scales and application scenarios, and reliability is evaluated from multiple aspects, including testing methods and failure mechanisms. This review aims to offer a clearer understanding of CPB technology and its future development.

2. Fabrication of copper pillar bump

The fabrication of CPBs mainly involves two key steps: first, copper pillars are formed on the under-bump metallization (UBM) layer of the chip through electroplating; subsequently, bumps are created on the top of the copper pillars. As the core interconnection structure between heterogeneous materials, the bump plays a decisive role in determining the mechanical strength, electrical performance, and long-term reliability of the package. Currently, the materials and fabrication methods for bumps mainly include electroplated solder [24–27], dip-transformation of copper pastes [28–32], deposition of Ag films

[33–35], electroplated Cu-Zn alloy followed by dealloying [36–39], and pulsed-laser deposition of nanoparticles [40].

2.1. Electroplating of solder bumps

The electroplating method for fabricating CPBs offers advantages such as a simple process, ease of mass production, and high stability [25, 41], making it widely adopted in commercial applications [42]. The fabrication process involves the following steps [43] (Fig. 3A): First, the wafer surface is coated with polyimide (PI), which is then exposed and developed to define openings that expose the pads. Then, Ti or TiW and Cu seed layers are sputtered to form the UBM. Next, photolithography is employed to define the pattern, followed by electroplating to form the copper pillars and the solder on top. Finally, the photoresist is stripped, the excess UBM is etched, and a reflow process is performed to transform the top solder into hemispherical bumps. Fig. 3B presents the morphology of the CPB array [44], whose performance is primarily determined by the composition and microstructure of the solder. Pure Sn was widely used as an early bump material due to its simple electroplating process and low cost; however, it suffers from a high melting point (232 °C), poor wettability, and low mechanical strength [45]. To optimize bump performance, the industry introduced Pb to form Sn-Pb eutectic alloys (melting point of 183 °C), which not only reduce interconnection temperatures but also significantly improve the wettability between the bump and substrate [46]. However, with the enforcement of the EU RoHS Directive (2002/95/EC), which restricts Pb-containing materials, bump compositions have shifted toward Sn-Ag alloys (Fig. 3C) [47,48]. The addition of Ag increases the tensile strength of solder joints by ~30% and significantly extends the high-temperature creep life of interconnects through solid-solution strengthening [49]. However, studies have shown that when the Ag content exceeds 3.5 wt %, needle-like Ag₃Sn precipitates form and increase the risk of interfacial brittle fracture [50,51]. On this basis, the SnAgCu (SAC) ternary solder alloy—optimized by adding 0.5–0.7 wt% Cu to improve mechanical performance—has gradually become the mainstream electroplated bump material due to its excellent resistance to electromigration and enhanced mechanical strength [52,53]. Moreover, in fields such as optoelectronic packaging and aerospace devices—where high service temperatures and extreme-condition reliability are required—the eutectic Au₈₀Sn₂₀ solder (melting point of 280 °C) has become a key bump material (Fig. 3D), owing to its high melting point, excellent fatigue resistance, and flux-free processing advantages [54,55].

In addition to solder composition, the reflow process influences the quality of bumps. During reflow, the solder is heated until it melts, and the molten solder spontaneously contracts into a hemispherical shape under the action of surface tension, thereby eliminating angular residues

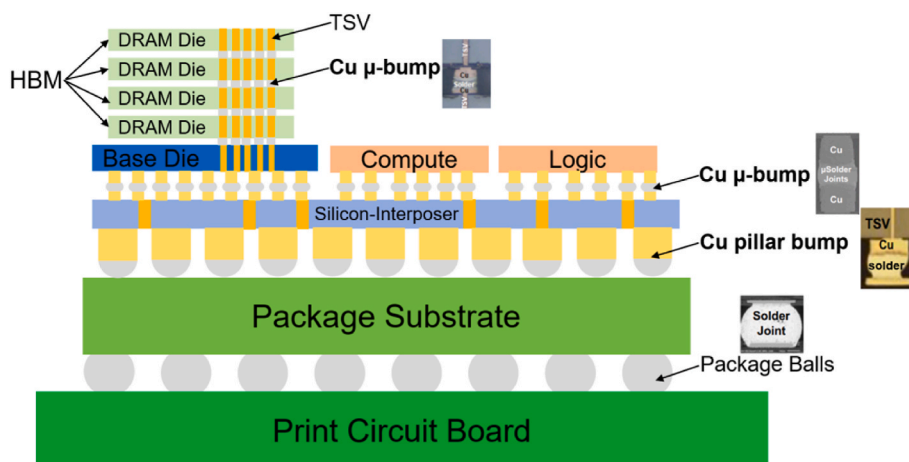


Fig. 1. Schematic diagram of a 2.5D packaging structure.

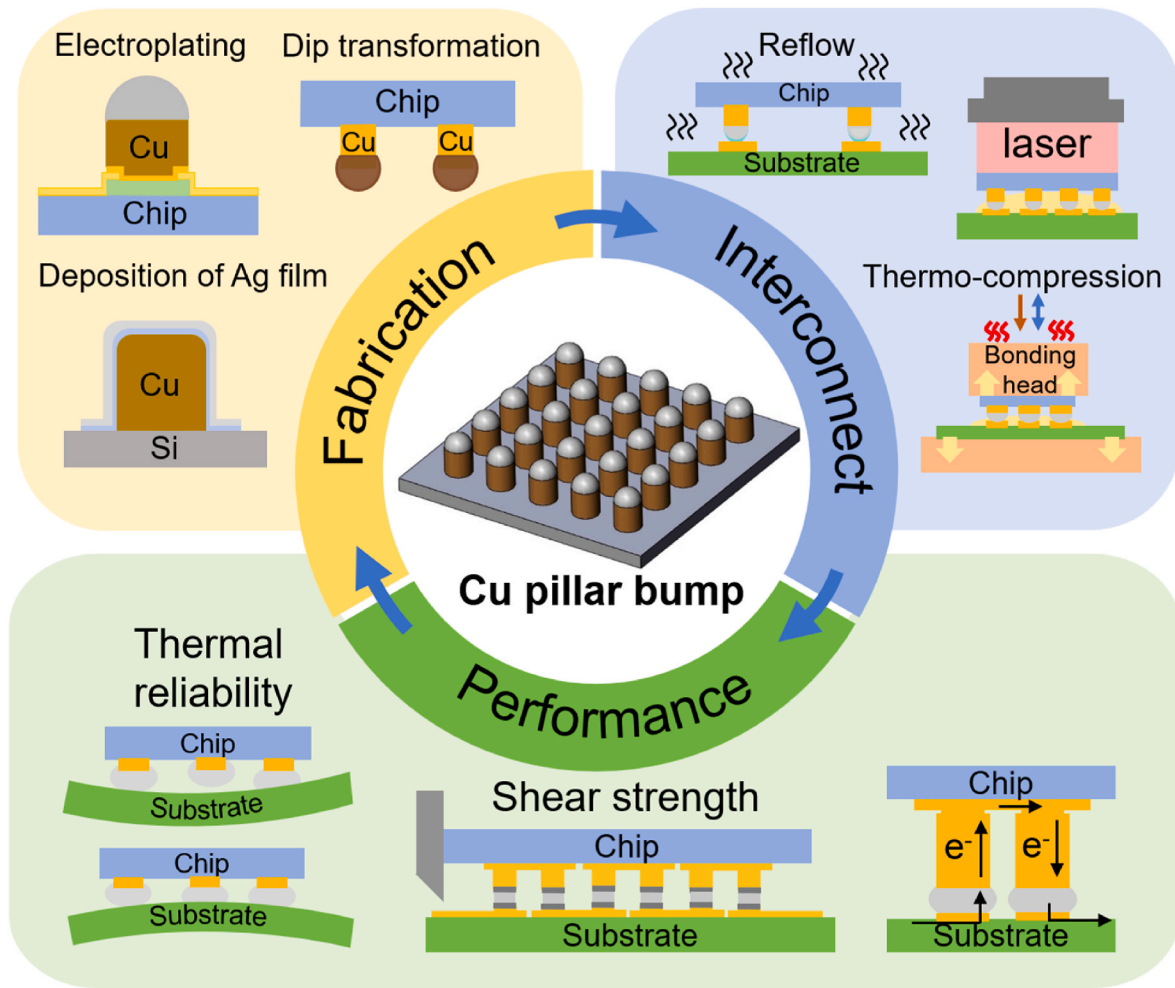


Fig. 2. Cu pillar bump technology for high-density electronic packaging.

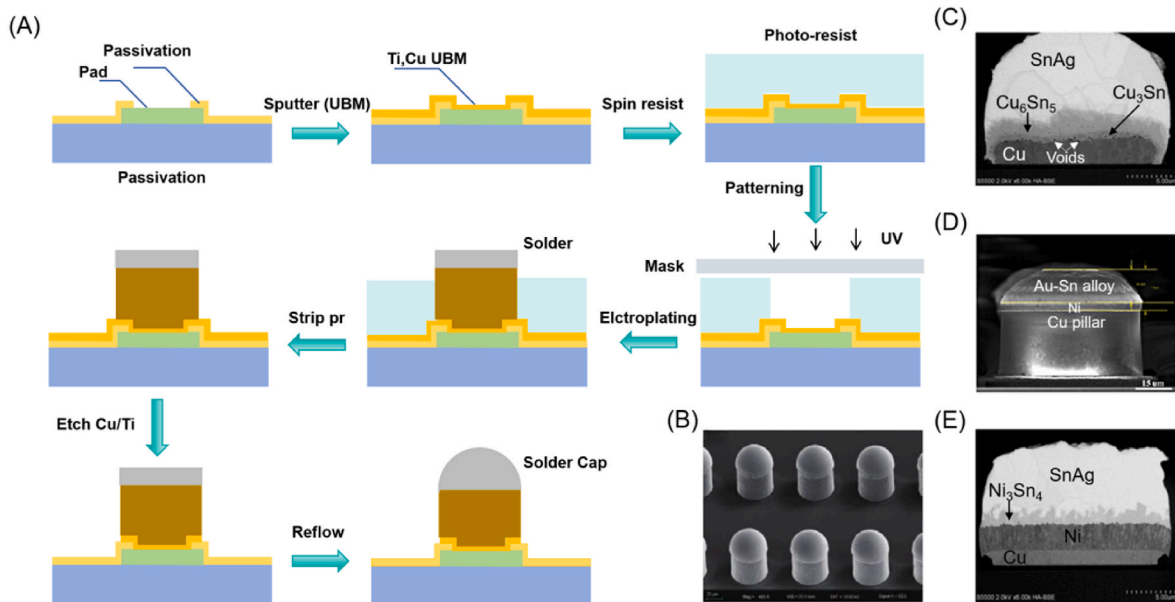


Fig. 3. Electroplating process and microstructures of solder bumps. (A) Schematic diagram of the electroplating-reflow process [43]. (B) Image of Cu pillar bump array. Reproduced with permission from Ref. [44]. Copyright 2018, Science Press. (C) Microstructure of SnAg bumps. Reproduced with permission from Ref. [48]. Copyright 2017, IEEE. (D) Microstructure of AuSn bumps with a Ni barrier layer. Reproduced with permission from Ref. [54]. Copyright 2024, Springer Nature. (E) Microstructure of SnAg bumps with a Ni barrier layer. Reproduced with permission from Ref. [48]. Copyright 2017, IEEE.

and irregular edges left from electroplating. However, during the reflow process, mutual diffusion occurs between the solder bump and the copper pillar, leading to the formation of intermetallic compounds (IMCs) such as Cu_3Sn and Cu_6Sn_5 (Fig. 3C) [56–58]. Because IMCs are inherently brittle and tend to accumulate microvoids, excessive IMC formation can significantly reduce the reliability of solder joints [59]. To mitigate IMC growth at the copper pillar/solder interface, researchers have proposed electroplating a Ni barrier layer between them (Fig. 3E), which forms Ni_3Sn_4 at the interface and effectively slows down the IMC formation rate [60]. Table 1 summarizes the properties of commonly used bump materials and their corresponding IMCs, enabling researchers to select and tailor materials according to specific application requirements.

Although the electroplating process offers advantages of stability and reliability in bump fabrication, traditional alloy solder systems require multiple high-temperature reflow cycles after electroplating, leading to risks such as excessive IMC growth and wafer warpage [68]. When the pitch is smaller than 40 μm , solder melting can also cause bridging between adjacent bumps [69], which limits the applicability of electroplating-reflow technology. In addition, the electrical conductivity of alloy solders is lower than that of pure metals, limiting the current-carrying capacity of the joints and posing challenges under high current densities [70].

2.2. Dip-transfer process of copper paste

To overcome the drawbacks of alloy solders—such as high bonding temperature, brittle IMC formation, and limited current-carrying capability—a technique employing pure copper pastes as the bump material to achieve all-copper interconnection has been proposed [28,71]. Owing to the size effect, nanoparticles can achieve sinter-joining at temperatures far below the melting point of the bulk material [72–74]. Compared with alloy solders, all-copper interconnection formed from copper pastes not only eliminate IMC formation but also exhibit higher current-carrying capacity and thermal conductivity [32,75]. Zürcher et al. [71] proposed a dip-transfer method to deposit copper pastes onto copper pillars. As shown in Fig. 4A, the copper pillars are first dipped into a copper pastes film of controlled thickness. Subsequently, the chip is withdrawn from the copper pastes at a controlled speed. During this process, the copper pastes adhere to the top of the copper pillars through the combined effects of capillary and surface tension forces, forming uniform copper paste bumps. Moreover, researchers have investigated the factors influencing the stability of the dip-transfer process and bonding quality, including withdrawal speed, copper pillar height, and the characteristics of the copper pastes. Carro et al. [30] found that within a withdrawal speed range of 0.5–4 mm/s, the thickness of the copper pastes coating on the pillar tips remained consistently between 11 and 13 μm , indicating that the pastes thickness is independent of the withdrawal speed. It is worth noting that the dip-transfer method can

form a relatively thick layer at the tips of the copper pillars, which can potentially compensate for height nonuniformities among the copper pillars (Fig. 4B and C).

Zürcher et al. [28] further investigated the effect of copper paste characteristics on bump joint strength. The results showed that bumps fabricated with pastes comprising solely copper nanoparticles exhibited continuous cracks and voids at the interconnection interface (Fig. 4D), achieving a shear strength of only 8 MPa at 250 °C. In contrast, bumps fabricated using bimodal pastes containing both nano- and micron-copper particles exhibited fewer cracks and voids at the sintered joint (Fig. 4E), achieving a shear strength of 19 MPa even at 200 °C. This behavior arises from the high reactivity of nanoparticles, which tend to agglomerate and form non-uniform packing during the sintering process. The incorporation of micron-copper particles provides a structural backbone, resulting in more uniform and dense particle packing. Fig. 4F and G shows the fracture surfaces of bumps sintered with different particle ratios, where the joint containing 40% μm -sized particles exhibits the fewest cracks. Recently, Liu et al. [31] successfully transferred copper pastes onto copper pillars with a diameter of 20 μm using the dip-transfer method. Thermocompression bonding was subsequently carried out in an N_2 atmosphere, achieving a shear strength of 31.87 MPa. SEM images of the sintered joints confirm that an optimal amount of copper pastes was transferred and a robust metallurgical bond was formed, further demonstrating the feasibility of the dip-transfer technique for fine-pitch interconnections.

However, insufficient transfer or overflow of copper pastes often occurs during the transfer process, which may result in open-circuit or short-circuit failures. Therefore, further optimization of the copper paste properties, transfer process, and sintering conditions are required to enhance the stability of the transfer method.

2.3. Deposition of Ag film

In recent years, studies have shown that depositing Ag on the Cu surface as an intermediate interconnection layer enables low-temperature diffusion bonding while effectively suppressing Cu oxidation [76,77]. Therefore, researchers have proposed using an Ag layer to achieve low-temperature bonding for CPBs. For example, Wang et al. [35] deposited a 3 μm Ag layer on the surface of Cu pillars via magnetron sputtering, followed by oxygen plasma treatment to induce the growth of Ag_2O nanoparticles on the Ag surface (Fig. 5A–C). During bonding, a two-step mechanism is involved. In the pre-bonding stage, the excellent plastic deformation capability of Ag enables the Ag layer to accommodate surface roughness and ensure sufficient contact between micro-bumps. More importantly, in the subsequent bonding stage at approximately 220 °C, the Ag_2O nanoparticles decompose in air, releasing highly active Ag atoms that significantly enhance surface diffusion and atomic rearrangement at the bonding interface. This decomposition-induced activation plays a critical role in enabling

Table 1
Physical properties of common alloy solders and IMCs.

Composition	Material	Shear modulus (GPa)	Shear strength (MPa)	Thermal conductivity (W/m•K)	Electrical resistivity ($\mu\Omega\cdot\text{cm}$)	Melting point (°C)	Ref.
Solder	Sn	18	20	66	11.5	232	[52,61, 62]
	$\text{Sn}_{63}\text{Pb}_{37}$	15	26	47	15	183	[63]
	$\text{SnAg}_{3.5}$	19	31.7	60	12–15	221	[63]
	$\text{SnAg}_{3.0}\text{Cu}_{0.5}$	20	40–50	61	12–15	217	[63]
	$\text{Au}_{80}\text{Sn}_{20}$	30	45–50	57	22	280	[64,65]
IMC	Cu_6Sn_5	37	670	34.1	17.5	415	[61,62, 66]
	Cu_3Sn	46.	1700–4400	70.4	8.8	676	[61,62, 66]
	Ag_3Sn	35	150–200	N/A	5–7	480	[66,67]
	Ni_3Sn_4	56	600–900	19.6	28.5	795	[66,67]

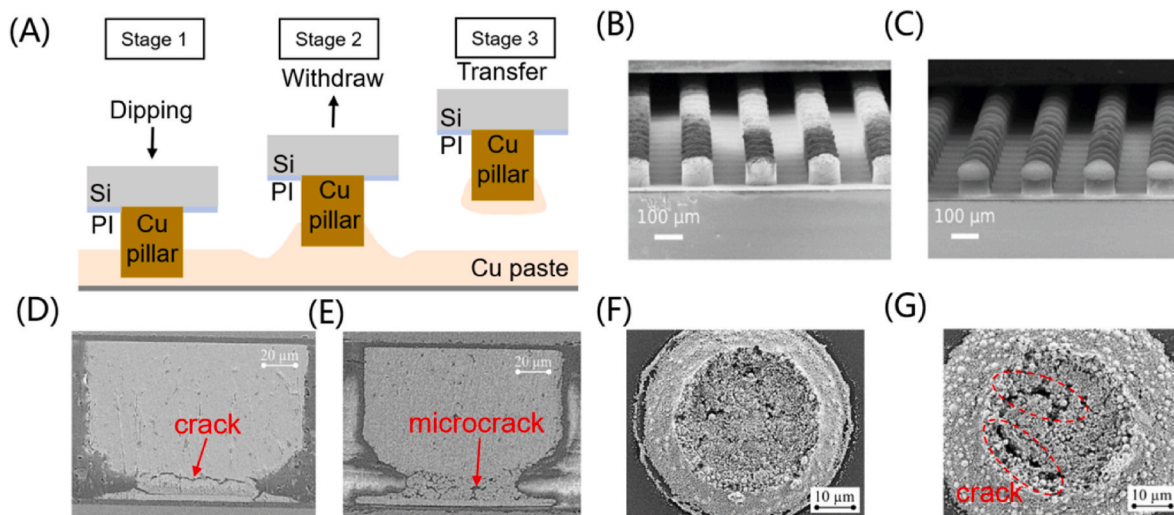


Fig. 4. Dip-transfer process and corresponding characterizations. (A) Schematic of dip-transfer process. Morphology of Cu pillars (110 μm in diameter and 220 μm in pitch) (B) before and (C) after dip-transfer, respectively. (B–C) Reproduced with permission from Ref. [30]. Copyright 2019, IEEE. (D) Cross-sectional image of an all-Cu joint formed using nanoparticle pastes. (E) Cross-sectional image of an all-Cu joint formed using mixed nano-and micron-particle pastes. Fracture surfaces of joints with (F) 40% and (G) 60% micron-particles. (D–G) Reproduced with permission from Ref. [28]. Copyright 2016, IEEE.

effective metallurgical bonding at relatively low temperature, leading to the formation of continuous Ag grains across the interface. As a result, a shear strength of 49.9 MPa was achieved under a bonding pressure of 20 MPa. On the other hand, Zhang et al. [34] deposited a Ti/Ag layer on the Cu pillar surface via RF sputtering, enabling full coverage of the Cu pillars with Ag (Fig. 5D). Fig. 5E shows the Cu pillar array capped with Ag. The Ag coating on the Cu pillar surface was bonded with the Ag caps on the substrate (Fig. 5F). Abnormal grain growth of Ag can induce hillock formation, which provides an ideal structure for filling gaps at the bonding interface [78]. In addition, Ag exhibits a high surface diffusion rate in the presence of oxygen [79], enabling a relatively flat bonding interface even when the top surface of the copper pillar is uneven (Fig. 5G). Bonding at 200 °C under a pressure of 5 MPa for 3 min resulted in a shear strength of 144.1 MPa.

Depositing metals on copper surfaces as anti-oxidation layers and interconnection mediators for low-temperature bonding has been extensively studied [80,81]. However, there are relatively few studies directly applying this approach to CPBs. Moreover, since the diffusion rate of Ag in nitrogen is lower than in oxygen, the performance of the CPBs joints in industrial nitrogen-protected environments need to be further evaluated.

2.4. Other methods

In addition to the electroplating, dipping, and Ag film deposition methods mentioned above, researchers have also proposed innovative techniques for fabricating bumps. For example, Sosa et al. [39] developed a method to fabricate nanoporous Cu bumps by first electroplating a Cu–Zn alloy layer onto Cu pillars as a precursor and then selectively removing Zn through dealloying to form nanoporous Cu structures (Fig. 6A–C). Owing to its highly interconnected porous structure with large specific surface area and high surface energy, nanoporous Cu exhibits enhanced atomic diffusion and high surface activity, which facilitates bonding. During bonding, the nanoporous Cu can undergo pressure-less or low-pressure sintering to form dense Cu–Cu interconnections without introducing intermetallic compounds. In addition, oxidation of Cu during bonding is typically mitigated by using a reducing atmosphere, such as forming gas or formic acid, which helps remove surface oxides and maintain active Cu surfaces for effective bonding. As a result, the interconnection can operate for up to 1000 h under a high current density of 10^5 A/cm^2 without failure. On the other

hand, pulsed laser deposition (PLD) is a technique in which high-energy laser pulses strike the surface of a target material, causing it to evaporate and generate plasma, which subsequently deposits as a nanoscale film or particles on a substrate [82]. Wang et al. [40] employed PLD technology to evaporate Cu from a target and deposit it onto a substrate (Fig. 6D), forming an array of nanoparticle Cu bumps with a 120 μm pitch (Fig. 6E). The resulting bumps exhibited a low elastic modulus and could deform on the micrometer scale in the vertical direction (Fig. 6F), helping to alleviate stress concentration caused by surface unevenness during bonding and improving interfacial contact reliability. Fig. 6G shows a cross-section of the interface after sintering, and a strength of 20 MPa was achieved at 160 °C (Fig. 6H). However, strict requirements on the photoresist profile and particle deposition thickness increase process complexity, and further investigation is needed into the electrical properties and fine-pitch scalability of the interconnections.

In addition to electroplating-based fabrication, various alternative transfer and patterning strategies have been explored for CPB formation. These approaches can be broadly categorized into dip-based transfer, deposition-assisted methods, and lift-off-assisted patterning techniques. Dip-based methods, such as Cu paste or particle transfer, are generally simple and compatible with large-area processing, making them attractive for scalable fabrication. However, their pattern definition is relatively limited, and achieving ultra-fine-pitch interconnections remains challenging due to material spreading and placement accuracy. Deposition of Ag film, offer improved uniformity and interface quality. Nevertheless, they typically require additional patterning processes to define bump geometries, which increases process complexity. Lift-off-assisted methods, for example those combining pulsed laser deposition (PLD) with lithographic patterning, enable precise control of bump geometry and high-resolution patterning, making them suitable for fine-pitch interconnections. However, these approaches rely on more complex fabrication steps and are less favorable for large-scale manufacturing. Overall, these methods exhibit trade-offs among scalability, patterning resolution, and process complexity, and their selection depends on the specific requirements of interconnection pitch and application scenarios.

3. Interconnection methods for copper pillar bumps

CPB interconnection refers to achieving high-density and reliable electrical and mechanical connections between a chip and a substrate, or

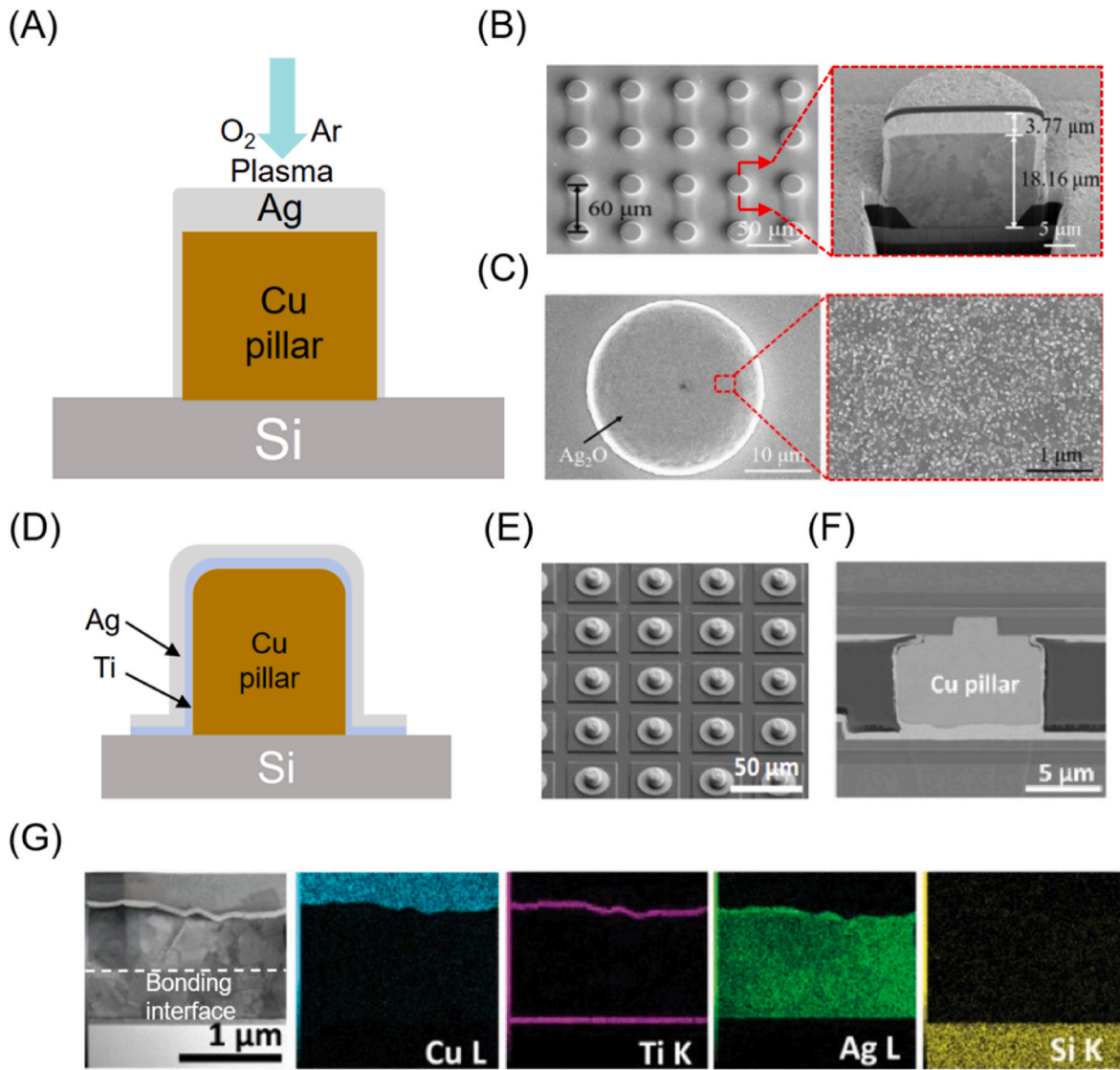


Fig. 5. Cu pillar bumps with Ag coated layers. (A) Schematic of Cu pillar bumps with Ag and plasma-induced Ag_2O layer. (B) Surface morphology and cross-sectional image of bumps. (C) Surface morphology of Ag_2O layer. (B-C) Reproduced with permission from Ref. [35]. Copyright 2024, IEEE. (D) Cu pillar bump with an Ag capping structure. (E) Morphology of the Cu pillar bump array. (F) Cross-sectional image of a bonded sample with a 10 μm pitch. (G) EDS mapping results of the bonding interface region. (E-G) Reproduced with permission from Ref. [34]. Copyright 2025, IEEE.

between chips, through bumps [83]. Currently, the main interconnection methods include reflow [17], thermo-compression bonding (TCB) [84,85], and laser-assisted bonding (LAB) [86–88].

3.1. Reflow

Reflow soldering is a method in which a specific temperature profile is applied in a reflow oven to melt the solder, allowing surface tension and self-alignment effects to ensure precise alignment and reliable interconnection between the CPBs and the pads. As shown in Fig. 7A [89], the process begins with dipping the bumps in flux, followed by aligning the chip and substrate, performing reflow soldering, and finally removing the flux residue. In addition, the selection of flux plays a critical role in reflow soldering, particularly for fine-pitch applications. Common flux types include rosin-based, water-soluble, and no-clean fluxes, each with different cleaning requirements and residue characteristics. For ultra-fine pitch CPBs, flux residues may become trapped between closely spaced bumps, posing challenges for post-reflow cleaning and potentially affecting long-term reliability. Therefore, optimizing flux composition and cleaning processes is essential for

achieving high-quality interconnections in fine-pitch packaging. Fig. 7B illustrates a typical temperature profile used in reflow soldering [90]. The reflow process mainly consists of the preheating, soaking, reflow, and cooling stages. In the preheating stage, the chip and substrate are uniformly heated to prevent thermal stress damage, while the flux is activated and solvents evaporate, reducing spattering and void formation during soldering. During the soaking stage, a stable temperature is maintained to eliminate temperature differences among components, ensuring overall thermal equilibrium; at this stage, the flux further cleans the pad and bump surfaces by removing oxides. In the reflow stage, molten solder wets the substrate and interacts at the interface [91]. In the cooling stage, the cooling rate is controlled to solidify the molten solder and form robust solder joints.

A thin and continuous IMC at the interface is essential for achieving reliable bonding [93,94]. It is worth noting that IMCs are brittle; excessively large or thick IMCs weaken the connection between the solder and metal layers and tend to induce microcracks within the joint. Fig. 7C illustrates the bonding interfaces after different reflow durations, showing that the IMC layer grows thicker with longer time in the reflow zone [92]. In the reflow soldering process, the IMC thickness is primarily

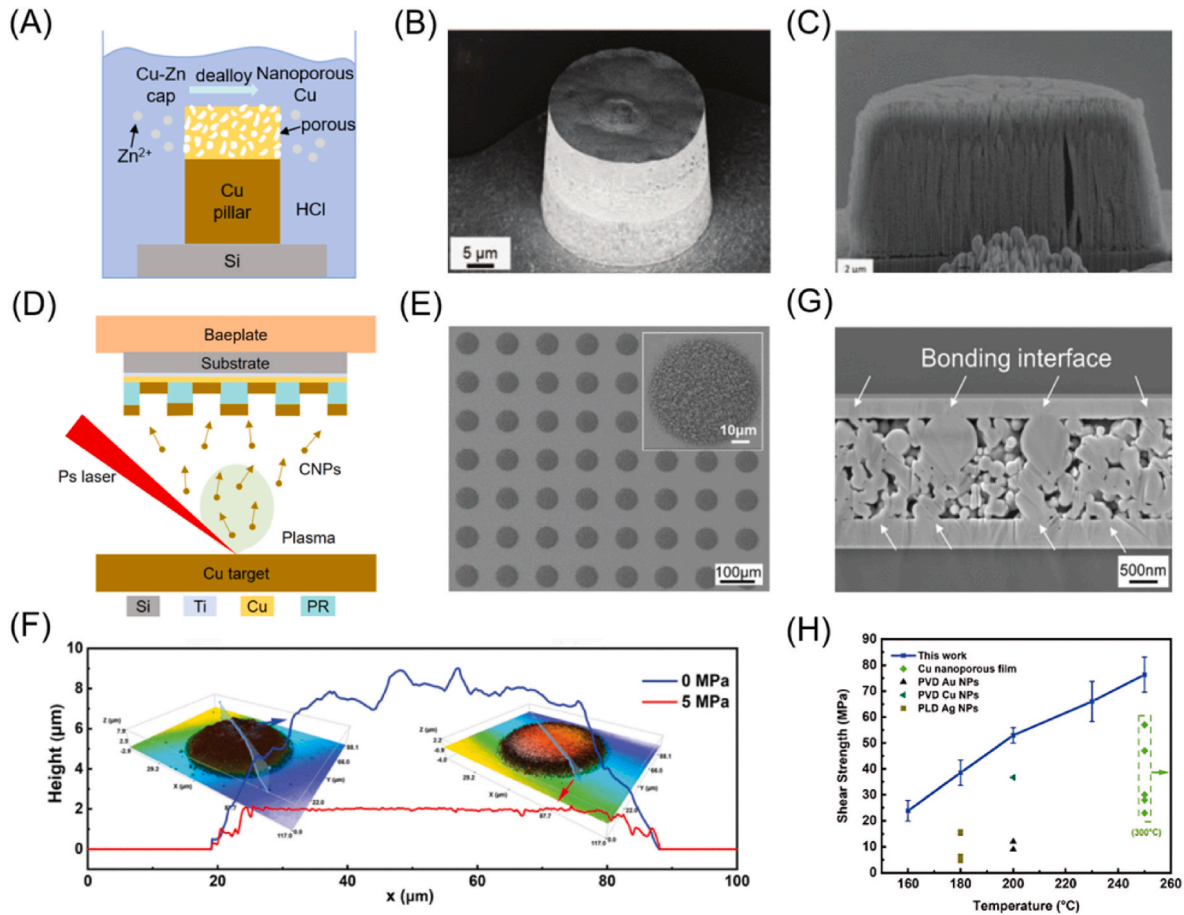


Fig. 6. Characterization of bumps prepared by dealloying and PLD. (A) Schematic of nanoporous Cu bumps fabricated via Cu-Zn dealloying. (B) Co-deposited Cu-Zn alloy on Cu pillars. (C) FIB cross-sectional image of nanoporous Cu bumps after dealloying. (B-C) Reproduced with permission from Ref. [39]. Copyright 2021, IEEE. (D) Schematic of bump fabrication by pulsed laser deposition (PLD). (E) Array of bumps prepared by PLD. (F) Profiles of bumps before and after compression. (G) Cross-sectional image of the bonded sample. (H) Variation of shear strength with temperature. (E-H) Reproduced with permission from Ref. [40]. Copyright 2023, IEEE.

influenced by the peak temperature and the time spent in the reflow zone. Wan et al. [95] optimized the reflow profile for bumps and found that increasing the peak temperature enhances the dissolution rate of copper in the molten solder, resulting in a thicker IMC. Similarly, extending the reflow duration increases the interaction time between copper and molten solder, leading to greater IMC growth. The cooling stage, where the molten solder solidifies to form solder joints, influences the microstructural evolution of solder joints (e.g., grain size, intermetallic compound growth, and internal stress), which in turn affects their mechanical properties such as tensile and shear strength. Lee et al. [96] conducted simulation studies to examine the influence of different copper pillar heights on temperature distribution, deformation, and thermal stress during the reflow process. The results indicated that a pillar height of 0.09 mm yielded the lowest reflow temperature, minimum peak deformation, and reduced thermal stress. At this optimal height, the team further investigated the effects of three bump materials—SAC305, SAC405, and SnBi—on the overall structure, and found that joints made with SAC305 exhibited the lowest von Mises stress. In summary, reflow quality is closely linked to precise control of the reflow profile, copper pillar height, and bump material selection, warranting continued investigation.

Although reflow soldering is well-suited for high-volume interconnect fabrication, shrinking copper pillar pitches and increasingly thinner chips and substrates have introduced significant process-stability challenges. As summarized in Fig. 7D, common reflow defects arise due to reduced self-alignment capability and differences in thermal expansion

coefficients, which likely generate thermal-gradient-induced warpage or tilt, leading to weak joints or voids [89,97]. For fine pitch bumps, solder bridging may occur during melting. In addition, without tight control of chip-substrate alignment, the bond-line spacing tends to fluctuate. Currently, reflow soldering remains viable for bumps with pitches above 40 μm; however, for smaller bumps and tighter pitches, reflow soldering is no longer suitable [95].

3.2. Thermo-compression bonding

To address solder bridging at fine pitches and warpage issues during interconnection, Intel pioneered the adoption of TCB to replace traditional reflow soldering. In the TCB process, solder is partially melted under simultaneous temperature and pressure application, and reliable joints are formed upon cooling, thereby mitigating the risk of bridging associated with complete solder melting. Additionally, TCB employs high-precision vacuum handling and Z-axis displacement control modules, which effectively reduce warpage during bonding and allow precise control over bump morphology. The primary steps and control parameters of the TCB process are illustrated in Fig. 8A and B [98]. First, the substrate is secured on the stage and heated to pre-activate the flux. Next, the bonding head picks up the chip and achieves alignment and contact with the substrate through optical recognition. The bonding head is then heated to partially melt the solder and held at temperature for a specified duration to promote interfacial reactions at each joint, corresponding to the heating and wetting stages in the process profile.

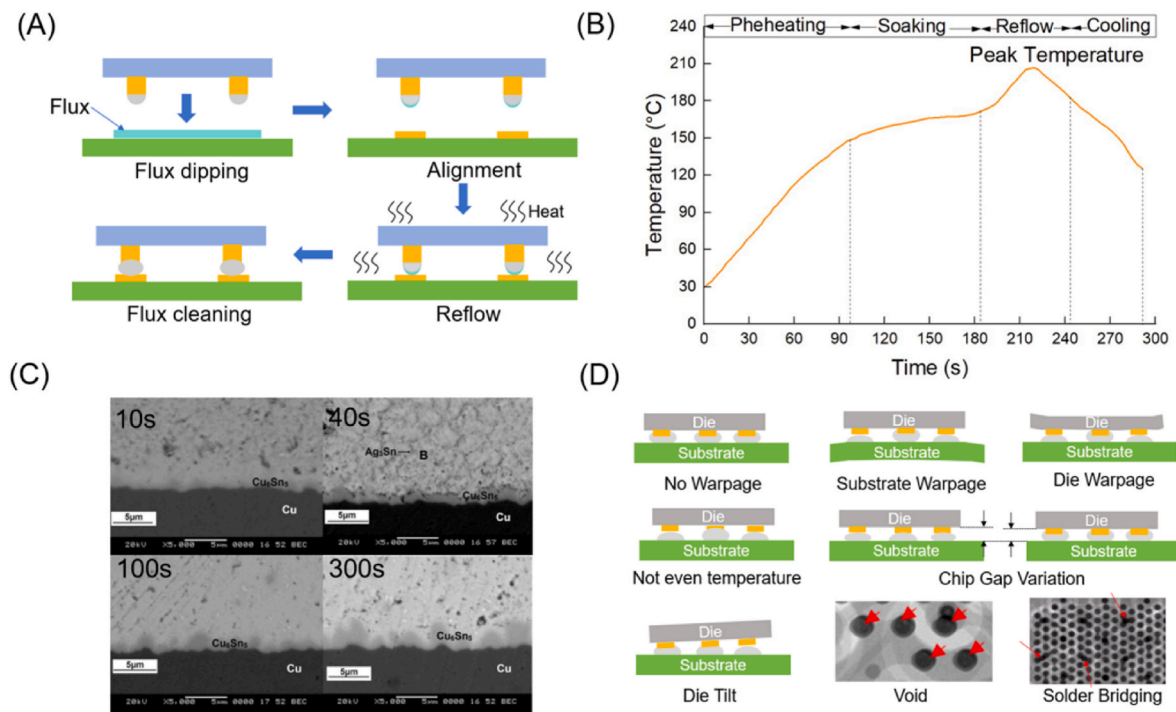


Fig. 7. Reflow soldering of Cu pillar bumps. (A) Process flow diagram of reflow soldering. (B) Typical reflow temperature profile. Reproduced with permission from Ref. [90]. Copyright 2009, Taylor & Francis. (C) Bonding interfaces after different reflow zone durations. Reproduced with permission from Ref. [92]. Copyright 2016, Springer Nature. (D) Typical defect types observed after reflow soldering. Reproduced with permission from Ref. [89]. Copyright 2015, IEEE.

Near the end of the wetting stage, displacement control can be applied to adjust the bump shape. Finally, during cooling, the joints return to room temperature, and the bonding head retracts and releases the chip.

The key to TCB lies in precise control of temperature, pressure, Z-axis displacement, and time, enabling effective adjustment of solder joint height and geometry. Fang et al. [100,99] systematically investigated temperature optimization in forming different bump morphologies using TCB. Tensile-type joints require higher bonding temperatures (350 °C) to reduce surface tension and internal stress, thereby preventing solder delamination (Fig. 8C). In contrast, compressive-type joints require lower temperatures (300 °C) to increase surface tension and avoid solder overflow (Fig. 8D). Huang et al. [101] investigated the impact of solder joint geometry on performance and reliability, and found that tensile-type joints exhibited the shortest thermal cycling lifetime, whereas compressive-type joints demonstrated superior electromigration resistance. Beyond joint geometry, the interfacial IMCs also affect interconnection reliability. Lu et al. [102] reported that IMC thickness increases with both time and temperature, with temperature playing a more dominant role. Excessively high temperatures cause excessive IMC growth, increasing the likelihood of cracking and leading to premature failure.

Copper paste-based bumps differ from solder bumps in that interconnection is achieved through solid-state sinter-joining rather than melting-solidification. This process enables particle bonding and the formation of dense joints. For example, Carro et al. [30] sintered copper pastes at 200 °C in a formic acid atmosphere, obtaining joints with a shear strength of 19 MPa. However, in the absence of applied pressure, it is difficult to achieve full densification, resulting in relatively high porosity. To this end, the effect of applied pressure on sinter-joining performance was further investigated. Sinter-joining was carried out at 200 °C under various pressures, and when a pressure of 50 MPa was applied, the porosity decreased from 21.3% (no pressure) to 1.6%, while the maximum shear strength reached 50 MPa. To prevent potential damage to interconnects caused by the sustained high temperature and pressure, Sosa et al. [103] proposed a tack-bond method, in which

pressure is applied only during the heating stage and removed once the target temperature is reached. They studied the relationship between shear strength of joints, applied pressure, and holding time. The joint strength increased with pressure, and at 50 MPa, both 1-min and 5-min sintering times achieved shear strengths of 60–70 MPa, outperforming conventional TCB.

Although TCB enables fine-pitch interconnection, it suffers from thermal non-uniformity in both horizontal and vertical directions, which can lead to incomplete bonding or solder overflow. In addition, TCB is typically performed on individual chips, resulting in higher costs compared with reflow soldering. Copper paste bumps can achieve finer pitches and higher joint strength; however, they usually require the combined application of elevated temperature and high pressure.

3.3. Laser-assisted bonding

To address the warpage caused by the high thermal budget of reflow soldering and the low productivity of TCB, researchers have proposed LAB. In this method, a uniform square-shaped laser beam is used to irradiate the chip, enabling selective heating only in the interconnection region. This approach minimizes substrate thermal expansion and warpage. As illustrated in Fig. 9A, the process involves first coating a non-conductive film (NCF) on the substrate, followed by chip-to-substrate alignment, laser bonding, and finally NCF curing.

The key parameters of LAB are laser power and irradiation time. Han et al. [104] achieved interconnection of SnAg bumps using LAB technology. When the laser power density was 2.65 W/mm² and the irradiation time was 1.0 s, the resulting solder joints exhibited a shear strength of 22.69 MPa, comparable to those produced by TCB and reflow soldering (Fig. 9B). Fig. 9C presents the cross-sectional microstructure of the bump joints. Similar to reflow and TCB, LAB achieves reliable interconnection through the formation of interfacial IMCs. Studies have shown that the IMC thickness in LAB joints is comparable to that in TCB joints [106]. It is worth noting that excessive laser power density and irradiation time may lead to solder splashing (Fig. 9D), which is not a

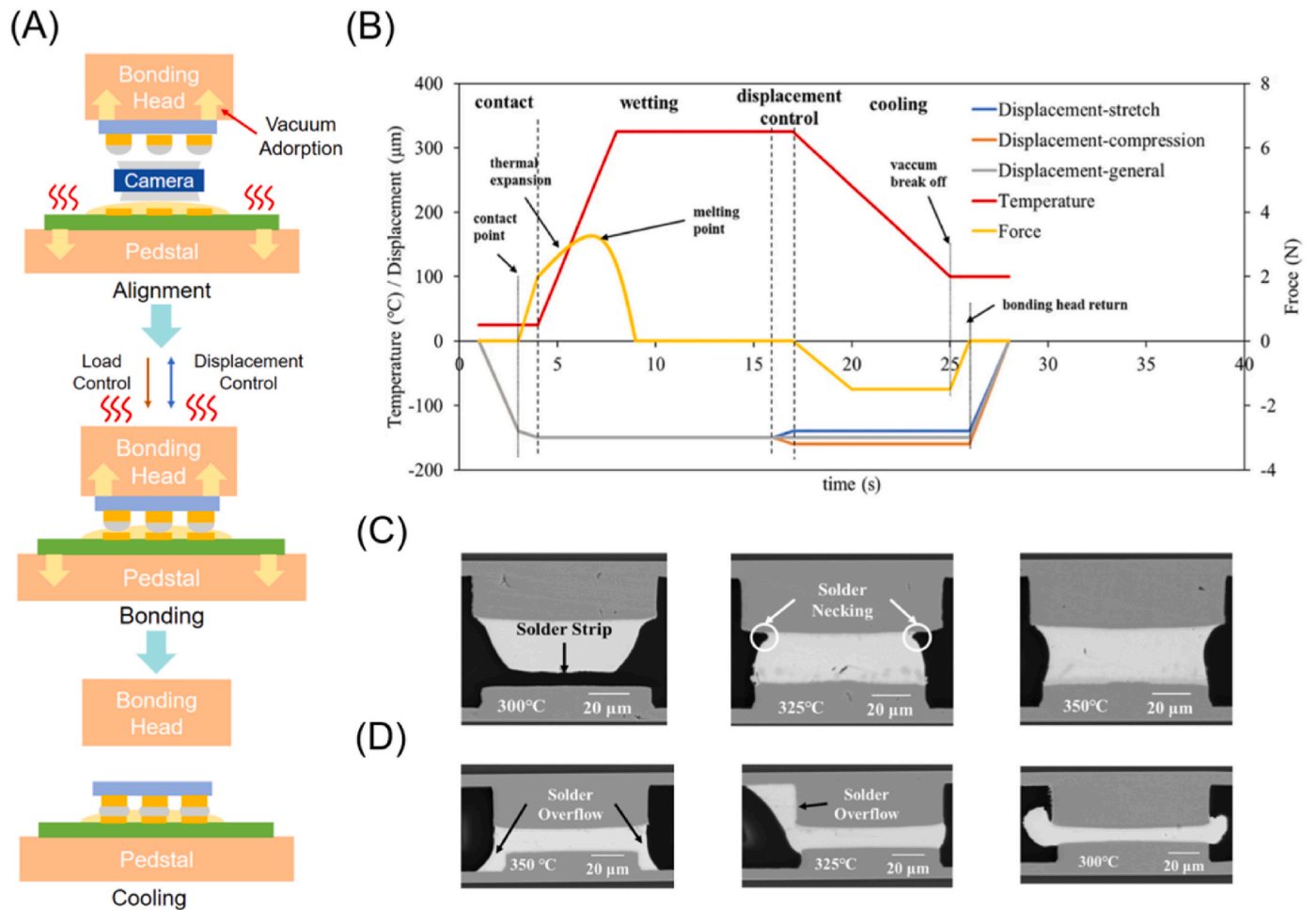


Fig. 8. Thermo-compression bonding and the effects of bonding parameters on solder joints. (A) Process flow diagram of thermo-compression bonding. (B) Parameter profile of the thermo-compression bonding. Morphology of (C) tensile-type and (D) compressive-type solder joints at different bonding temperatures, respectively. (B-D) Reproduced with permission from Ref. [99]. Copyright 2022, Springer Nature.

liquid-vapor phase transition of the solder itself, but rather results from the instability of molten solder under rapid heating, accompanied by thermal-stress buildup and confinement effects from the surrounding materials, ultimately causing a sharp decrease in shear strength. Jang et al. [88] further identified substrate thickness, the number of copper bumps, and bonding stage temperature as key factors influencing heat transfer. A thinner substrate, greater number of copper bumps, or lower stage temperature results in a lower wafer temperature, which tended to lead to incomplete solder melting.

LAB is also applicable to copper paste bumps. Carro et al. [105] found that the occurrence of sintering in the pastes primarily depends on the energy density—the product of irradiation intensity and duration. As shown in Fig. 9E, different energy densities lead to three distinct regions: unsintered, sintered, and oxidized. Fig. 9F presents the corresponding optical and SEM images of these regions. Within an appropriate energy density range (26–44 J/mm²), this method enables dense joints between copper paste bumps and the substrate (Fig. 9G), achieving an average shear strength of approximately 20 MPa. However, LAB technology is still in its early stage of development, and further research is needed to determine optimal laser parameters for achieving stable and reliable bump joints in production.

Accordingly, pitch scalability has become a key criterion for evaluating the applicability of different CPB interconnection methods in advanced packaging. Conventional reflow-based bonding is generally more suitable for relatively larger pitch interconnections (above 40 μm), because further pitch reduction is increasingly constrained by solder

bridging and package warpage. In addition, flux residue removal becomes more difficult as the pitch decreases, which may introduce additional reliability concerns. TCB is more favorable for fine-pitch applications due to its improved control of local bonding deformation. However, its pitch capability is still limited by factors such as thermal and pressure non-uniformity, alignment accuracy, and the compatibility of flux or underfill materials with the bonding process. LAB offers advantages in localized and rapid heating, which can help reduce the overall thermal budget and improve process efficiency, and has been explored for pitch scaling comparable to or below TCB. Nevertheless, its practical application is still sensitive to laser power, irradiation time, and process window control, which may affect bonding uniformity and stability at smaller pitch. In this context, throughput also becomes an important consideration, where reflow generally enables higher throughput for large-scale production, while TCB and LAB, although offering better process controllability for fine-pitch interconnections, typically operate at lower throughput.

4. Reliability of copper pillar bump joints

The mechanical, electrical, and thermomechanical properties of CPBs are key indicators of their long-term service reliability in advanced packaging. Conducting systematic reliability evaluations of these properties is an essential step toward transitioning from laboratory research to practical application. Currently, reliability testing is primarily focused on solder-based CPBs, including methods such as high-

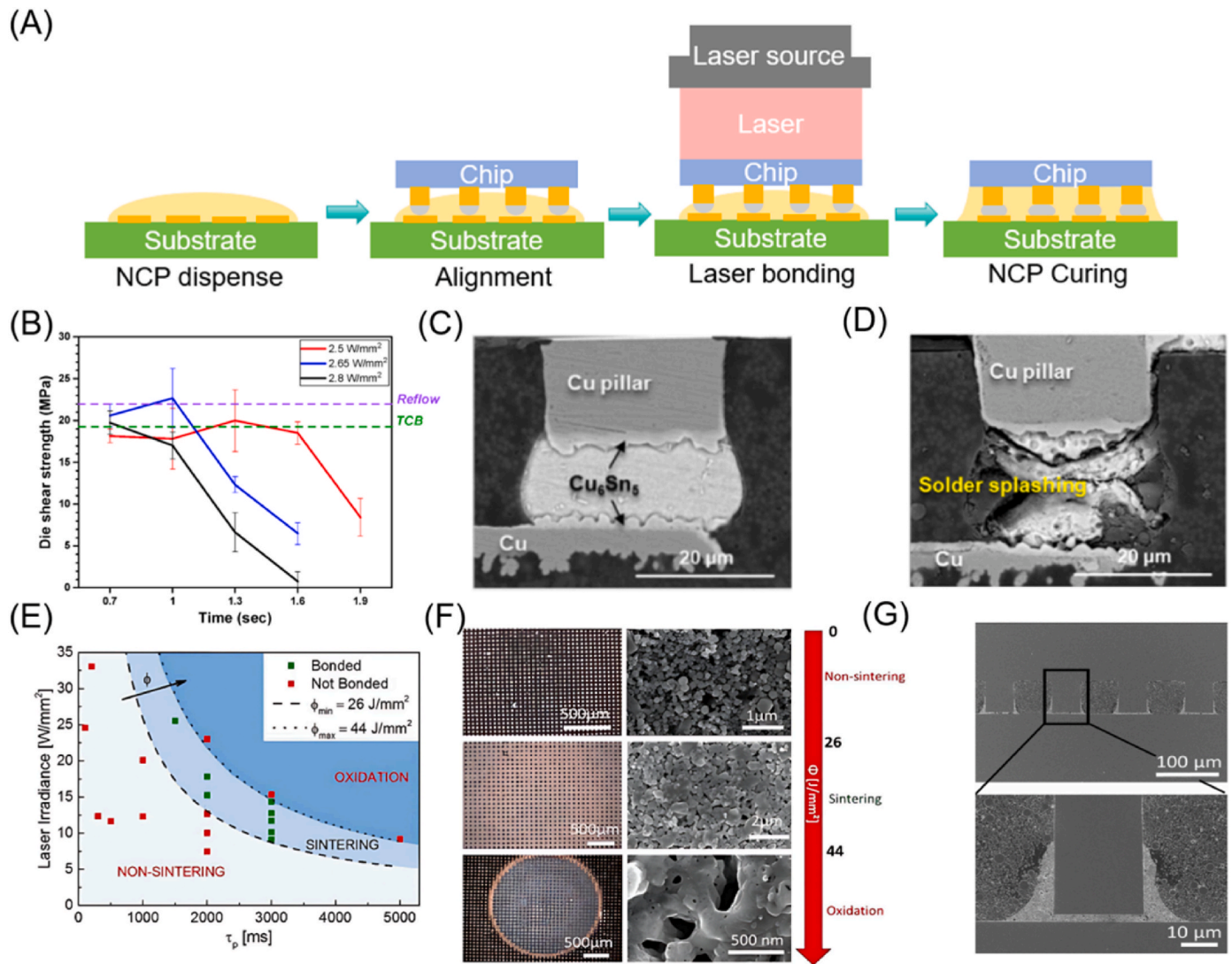


Fig. 9. Laser-assisted bonding of Cu pillar bumps. (A) Process flow diagram of the laser-assisted bonding method. (B) Variation of shear strength with bonding parameters. Cross-sectional image of the bump joint under bonding conditions of (C) 2.65 W/mm² for 1 s and (D) 2.65 W/mm² for 1.6 s, respectively. (B-D) Reproduced with permission from Ref. [104]. Copyright 2025, MDPI. (E) Relationship between the unsintered, sintered, and oxidized areas and laser parameters. (F) Bright-field optical images (left) and SEM images (right) of the unsintered, sintered, and oxidized areas. (G) Cross-sectional image of bump joints after laser-assisted bonding. (E-G) Reproduced with permission from Ref. [105]. Copyright 2018, IEEE.

temperature storage test (HTST), thermal cycling test (TCT)/thermal shock test (TST), and electromigration test (EM).

4.1. High temperature storage test

HTST involves aging the interconnect structure at a constant temperature to simulate the long-term operation of electronics under elevated temperatures. High temperatures accelerate IMC growth, promote microvoid formation, and cause thermal stress accumulation, all of which can degrade solder joint reliability. Detailed experimental procedures and standards are specified in JEDEC Standard JESD22-A103E.01 [107].

During high-temperature storage, void formation is a critical factor affecting the long-term reliability of CPB joints. These voids mainly originate from two distinct mechanisms: interfacial voiding associated with Cu–Sn interdiffusion and voiding within the residual solder related to mechanical effects during IMC growth.

The CPB joints are primarily composed of IMCs and a small amount of residual solder, or in some cases, entirely of IMCs [108]. Their microstructure tends to undergo phase transformation and evolution at

elevated temperatures. During this process, the different diffusion rates of Cu and Sn often result in the formation of excessive Kirkendall voids at the interface, leading to potential reliability concerns. Jing et al. [109] investigated the phase transformation and microstructural evolution of SnAg2.4 bumps during high-temperature storage at 160 °C. They found that the thickness of Cu₆Sn₅ remained relatively stable throughout the experiment, while the thickness of Cu₃Sn gradually increased due to the diffusion of Cu atoms through the Cu₆Sn₅ layer. This occurs because Cu₆Sn₅ acts as a diffusion barrier while the copper pillar continuously supplies Cu atoms to it (Fig. 10A). During the aging process, the shear strength of the joints first decreases and then increases (Fig. 10B). This behavior can be attributed to two main factors. First, the crystallographic orientation of Cu₆Sn₅ gradually changes from [011] to Ref. [110], resulting in increased brittleness. Second, uneven atomic diffusion between Cu₆Sn₅ and Cu₃Sn leads to the formation of Kirkendall voids at the interface, causing stress concentration and promoting crack initiation and propagation, which shifts the fracture mode from ductile to brittle. With continued aging, the shear strength increases after 30 days, likely due to local hardening of Cu₆Sn₅ and the precipitation of Ag₃Sn, which suppresses grain boundary sliding. This corresponds to a

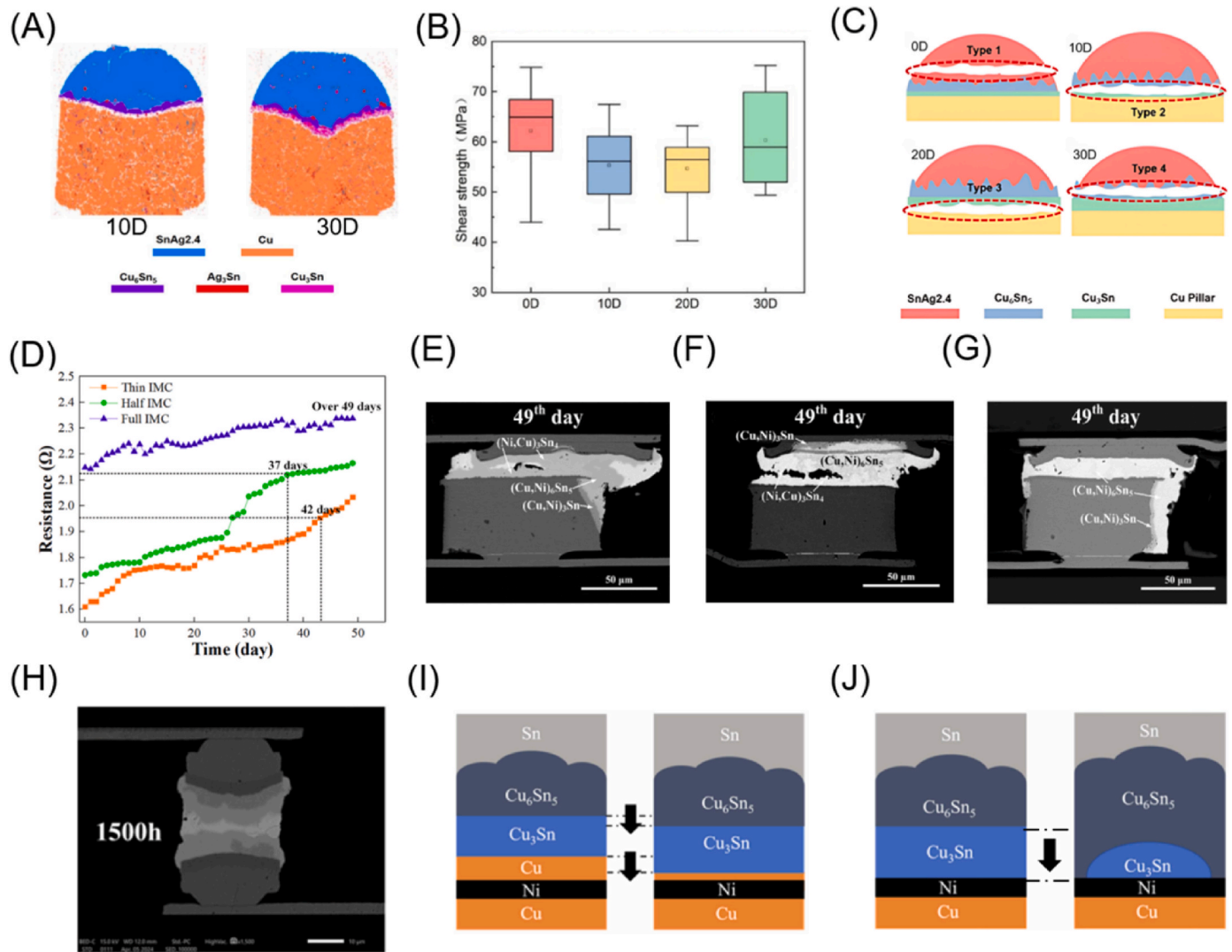


Fig. 10. High temperature storage test of Cu pillar bumps and joints. (A) Phase diagrams after 10 days and 30 days of testing. (B) Variation of shear strength with storage time. (C) Evolution of fracture modes. (A-C) Reproduced with permission from Ref. [109]. Copyright 2025, Elsevier Inc. (D) Change in joint resistance over time for different IMC contents. Microstructure of (E) thin-IMC, (F) semi-IMC and (G) full-IMC joints after 49 days, respectively. (D-G) Reproduced with permission from Ref. [111]. Copyright 2024, Elsevier B.V. (H) SEM image of the joint after 1500 h of thermal aging. (I) Diffusion model at the initial stage. (J) Diffusion model at the final stage. (H-J) Reproduced with permission from Ref. [112]. Copyright 2024, Elsevier B.V.

transition in fracture mode from intergranular to transgranular fracture (Fig. 10C). In addition, Wang et al. [111] found that the reliability of joints during HTST is closely related to the initial proportion of IMCs. As shown in Fig. 10D, the resistance of three types of joints changes with storage time: joints with thin IMC layers fail rapidly due to the gradual transformation of residual solder into IMCs accompanied by void formation; semi-IMC joints fail even earlier due to the smaller amount of solder; in contrast, full-IMC joints, despite their higher initial resistance, contain almost no residual solder and exhibit the most stable resistance behavior and superior reliability. As shown in Fig. 10E, after the experiment, the thin-IMC joints mainly consisted of a small portion of solder and a large amount of IMC, with significant void formation. The semi-IMC joints, after testing, were completely transformed into IMCs and exhibited extensive voids (Fig. 10F). In contrast, the full-IMC joints showed almost no change in void morphology after the experiment (Fig. 10G), consistent with their stable resistance behavior.

For interfacial voiding, although it is commonly described as Kirkendall voiding arising from the imbalance of Cu and Sn diffusion, its severity is strongly influenced by extrinsic factors. In particular, electroplating impurities can significantly promote void nucleation by

facilitating vacancy accumulation and lowering the energy barrier for void formation [113]. In addition, the subsequent growth and coalescence of these voids during thermal aging may be further affected by contamination from surrounding packaging materials, such as underfill.

By contrast, voiding within the solder is governed by the mechanical effects associated with IMC growth. As Cu–Sn IMCs form during aging, the accompanying volume shrinkage generates local tensile stresses in the residual solder, leading to plastic deformation and void formation [114]. This mechanism strongly depends on joint design parameters, including solder volume, solder height, solder geometry, and substrate stiffness. In general, sufficient solder volume and favorable geometry help relieve stress concentration, whereas insufficient solder or unfavorable structural design accelerates void formation.

Electroplating a Ni layer on the copper pillar can suppress the formation of Kirkendall voids; however, the interfacial Ni₃Sn₄ formed at the joint exhibits much lower electrical and thermal conductivity than Cu₃Sn and Cu₆Sn₅, and its higher brittleness significantly degrades the thermomechanical performance of the joint [115]. To address the above issue, Xu et al. [112] proposed a bump structure with an additional Cu layer electroplated on top of the Ni layer. The resulting joints exhibited

no noticeable voids after high-temperature storage at 150 °C for 1500 h (Fig. 10H) and successfully prevented the formation of Ni_3Sn_4 . This is because when the Cu layer in contact with Sn is relatively thin, the diffusion fluxes of Cu and Sn reach near equilibrium from the beginning (Fig. 10I). As Cu becomes depleted, the diffusion of Sn dominates, promoting the transformation of the IMC into Cu_6Sn_5 (Fig. 10J), thereby suppressing void formation and yielding a stable joint.

As an accelerated testing method, HTST can significantly accelerate chemical reactions and atomic diffusion within interconnect structures, allowing the long-term reliability to be predicted within a shorter timeframe. However, in real operating conditions, CPB joints are affected by multiple coupled factors, including material properties, size effects, and temperature gradients. Therefore, a single-temperature test cannot fully capture their complex failure behavior.

4.2. Thermal cycling/thermal shock test

Thermal cycling testing (TCT) subjects solder joints to alternating high and low temperatures to evaluate their ability to withstand thermally induced stress at component level (Fig. 11A). In practical flip-chip and micro-bump configurations, the thermo-mechanical stress is largely governed by the coefficient of thermal expansion (CTE) mismatch between the silicon chips, underfill materials, and solder joints, rather than only the mismatch among metallic components. During temperature cycling, the presence of underfill introduces additional constraint and redistributes the thermal strain, leading to significant stress accumulation within the solder and interfacial regions. Such nonuniform stress distributions can induce solder creep deformation and fatigue damage, ultimately resulting in crack initiation and propagation. Detailed testing procedures and standards can be found in the JEDEC Standard JESD22-A104F.01 [116].

Li et al. [117] compared the thermal cycling reliability of two types of CPB joints: thin-IMC and full-IMC. The results showed that thin-IMC joints exhibited longer thermal cycling lifetimes, with a plastic fracture mode in which cracks mainly initiated at the interface between the solder and IMC on the copper pillar side (Fig. 11B). In contrast, full-IMC joints failed through a brittle fracture mode, with cracks forming at the interface between Cu_6Sn_5 and Cu_3Sn (Fig. 11C). Stress concentration and crack propagation led to reduced thermal cycling lifetime. Chu et al. [118] further found that after thermal cycling, cracks in SnAg micro-bumps propagated not only along the solder/IMC interface (white arrows in Fig. 11D) but also along high-angle grain boundaries within the solder (red arrows in Fig. 11D and E). To improve the thermal cycling reliability of CPB joints, researchers have conducted studies on materials and processing optimization. For example, Shanawah et al. [121] found that SAC305 and SAC405 bumps exhibited significantly better thermal cycling performance than SAC105, owing to the presence of numerous Ag_3Sn particles and fine primary β -Sn grains, which suppressed plastic deformation and grain coarsening during high-temperature aging. Liu et al. [122] reported that adding $\alpha\text{-Fe}_2\text{O}_3$ nanoparticles improved the thermal cycling performance of the bumps; after 700 cycles, the joint strength remained around 50 MPa. This improvement was attributed to $\alpha\text{-Fe}_2\text{O}_3$ promoting grain refinement, adsorbing on grain boundaries to inhibit boundary sliding, and leading to a more uniform IMC layer with dispersion strengthening effects that enhanced joints strength.

Compared with TCT, TST involves much faster temperature changes, causing materials to undergo rapid thermal expansion and contraction within a short period of time, which more easily generates microcracks and thermal fatigue damage, thereby degrading performance (Fig. 11A). Detailed experimental procedures and standards are provided in JEDEC Standard JESD22-A106B.02 [110]. Chen et al. [119] investigated the effects of thermal shock on the mechanical properties and crystallographic evolution of SnAg2.4 bumps. The results showed that the shear strength decreased significantly with increasing thermal shock cycles (Fig. 11F). There are two main reasons for this phenomenon. First, as the

number of thermal shock increases, the average Sn grain area increases from $5.5 \mu\text{m}^2$ to $27 \mu\text{m}^2$ (Fig. 11G), weakening the grain refinement strengthening effect. Second, during thermal shock, Sn relieves part of the interfacial stress through creep, slip, and twinning; however, as the stress transfers toward the central region, it becomes difficult to dissipate outward, leading to stress accumulation and a reduction in strength (Fig. 11H). To enhance the thermal shock resistance of CPB joints, Wang et al. [120] fabricated $\text{Sn}_3\text{Ag-CoSn}_3$ composite bumps based on SnAg alloy. During the reflow process, CoSn_3 nanocrystals decomposed and generated numerous $(\text{Cu,Co})_6\text{Sn}_5$ nanoparticles within the SnAg_3 matrix (Fig. 11I). After 1500 thermal shock cycles, the composite joints showed no fracture, exhibiting only deformation in the UBM layer due to CTE mismatch (Fig. 11J). In contrast, pure SnAg joints experienced severe cracking caused by intergranular separation of the IMC layer at the interface (Fig. 11K). This is because the $(\text{Cu,Co})_6\text{Sn}_5$ nanoparticles strengthen the joints through a pinning effect and suppress excessive IMC growth.

Although CPBs exhibit good thermal reliability in thermal cycling and shock tests, single-temperature variations cannot accurately simulate the complex heat transfer conditions and environmental factors encountered in real applications. Therefore, further research is needed to evaluate the reliability of CPB joints under actual service conditions.

4.3. Electromigration test

EM test is an accelerated experiment conducted under various current magnitudes and directions to evaluate the resistance of CPB joints to electromigration-induced failure (Fig. 12A). Under the influence of electron wind generated by the flow of electrons, metal atoms gradually migrate, leading to the formation of vacancies and voids along their migration paths, or metal accumulation in certain regions [123]. The applied current accelerates microstructural evolution within the joints and even induces cracking at different interfaces. Detailed testing procedures and standards can be found in JEDEC Standard JEP154A [124].

The microstructural evolution of joints under current stressing is key to understanding the mechanism of electromigration failure. Yao et al. [125] found that the entire electromigration failure process can be divided into two stages (Fig. 12B). First, Cu and Sn interdiffuse to form Cu_3Sn , and Kirkendall voids develop at the Cu/ Cu_3Sn interface. At this stage, IMC growth and phase transformation have little effect on resistance. Subsequently, the voids coalesce under the electron wind force to form continuous cracks, which reduces the effective contact area, leading to a sharp increase in resistance and eventual failure.

Adding a Ni layer on the copper pillar can block Cu atom diffusion, thereby retarding IMC formation and improving electromigration reliability. For instance, Park [126] systematically studied the effect of Ni barrier layers on the electromigration failure mechanism of micro-bump structures with copper pillars on both sides. Since Cu diffusion into SnAg solder is limited by the Ni barrier layer and the interfacial Ni_3Sn_4 , the use of symmetric Ni barriers results in a significantly longer electromigration lifetime than that of asymmetric configurations (Fig. 12C). Introducing alloying elements into the solder is also an effective way to enhance electromigration resistance. For example, Zhao et al. [130] found that adding Ge to SAC305 bumps caused Ge atoms to segregate along grain boundaries, thereby preventing grain boundary sliding and suppressing Sn atom diffusion. Liu et al. [131] reported that adding 1 wt % Zn to SnAg1Cu0.5 bumps led to the formation of a stable Cu_5Zn_8 phase, which effectively hindered the dissolution of Cu_6Sn_5 on the cathode side. Meanwhile, the back diffusion of Zn filled the vacancies generated by Sn migration, alleviating vacancy accumulation and enabling the joints to remain stable under electromigration test. In addition, as the primary constituent of joints, the orientation of Sn grains is closely related to the diffusion of solute atoms under current stressing, thereby strongly influencing electromigration reliability. Xu et al. [127] investigated the effect of β -Sn grain orientation on the microstructural evolution of SnAg_{1.8} solder joints under conditions of $1.5 \times 10^4 \text{ A/cm}^2$

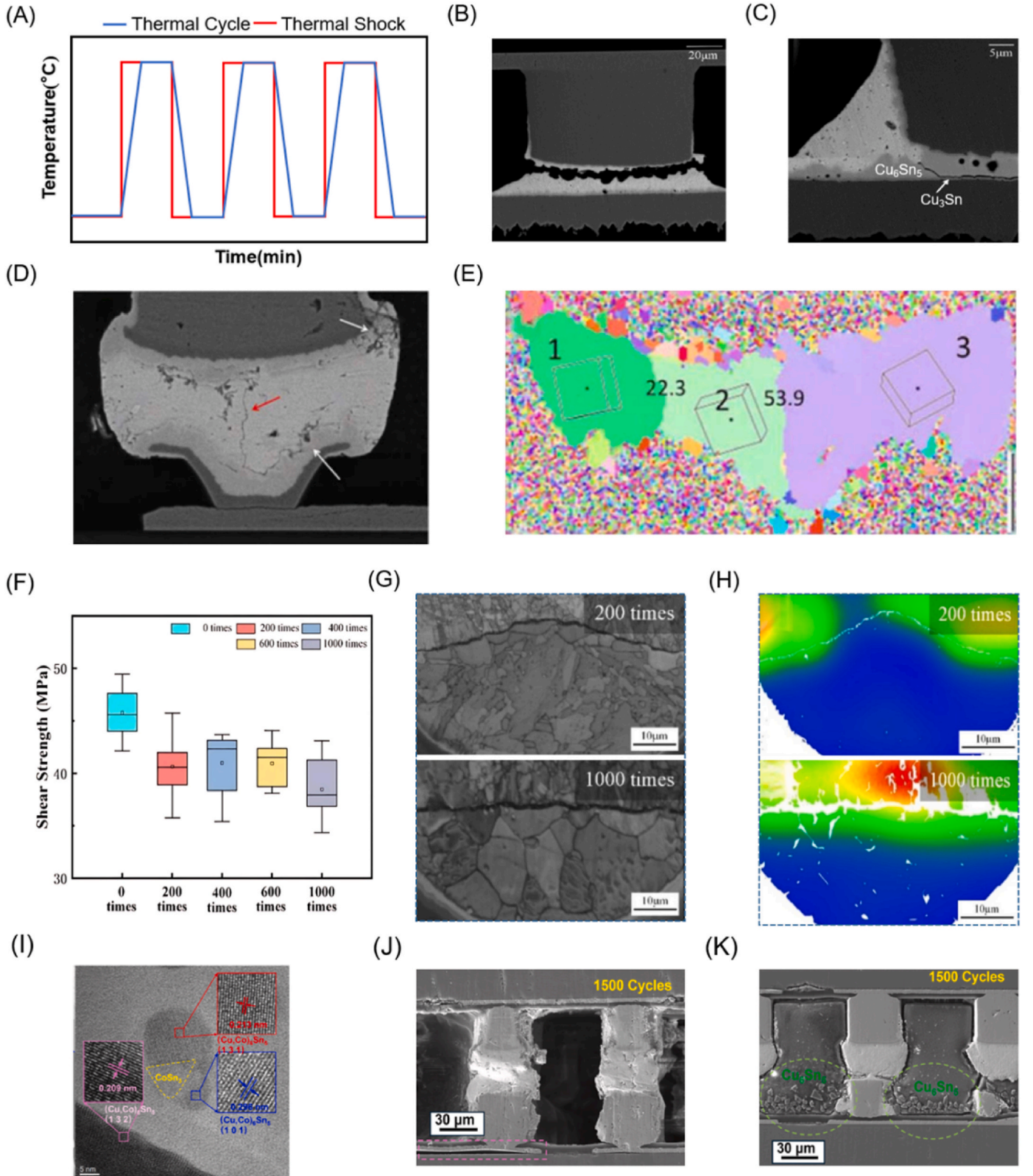


Fig. 11. Thermal cycling and thermal shock tests of Cu pillar bumps and joints. (A) Temperature profile of the thermal cycling/shock test. Fracture morphology of (B) thin-IMC and (C) full-IMC joint after thermal cycling test. (B-C) Reproduced with permission from Ref. [117]. Copyright 2020, Elsevier Ltd. (D) Cross-sectional image of micro-bumps after 2500 thermal cycles. (E) Corresponding OIM (Orientation Imaging Microscopy) map of (D). (D-E) Reproduced with permission from Ref. [118]. Copyright 2017, Springer Nature. (F) Variation of shear strength with the number of thermal shock cycles. (G) Grain evolution after different thermal shock cycles. (H) Stress distribution under different thermal shock cycles. (F-H) Reproduced with permission from Ref. [119]. Copyright 2025, Elsevier Ltd. (I) TEM image of $(\text{Cu},\text{Co})_6\text{Sn}_5$. Interfacial morphology of (J) $\text{Sn}_3\text{Ag}-\text{CoSn}_3$ and (K) Sn_3Ag joints after 1500 thermal shock cycles, respectively. (I-K) Reproduced with permission from Ref. [120]. Copyright 2024, Elsevier B.V.

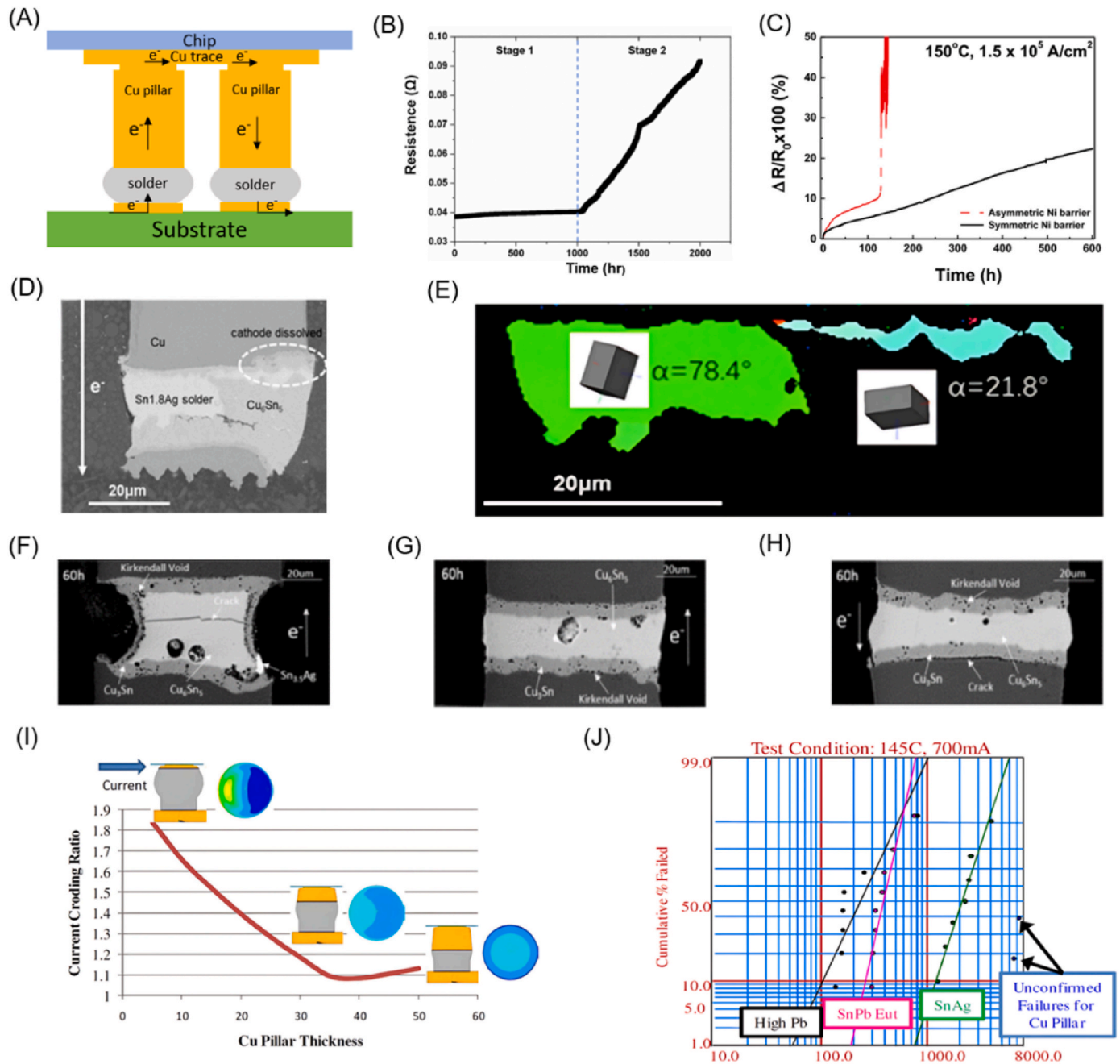


Fig. 12. Electromigration test of Cu pillar bump joints. (A) Common structure used for electromigration test. (B) Resistance evolution during the electromigration-induced failure process. Reproduced with permission from Ref. [125]. Copyright 2014, IEEE. (C) Variation of resistance with test time for symmetric and asymmetric Ni-layer micro-bump joints. Reproduced with permission from Ref. [126]. Copyright 2019, Springer Nature. (D) Microstructure of bump joints after electromigration test. (E) Corresponding electron backscatter diffraction (EBSD) map of (D). (D-E) Reproduced with permission from Ref. [127]. Copyright 2021, MDPI. Microstructures of (F) cylindrical (G) hourglass-shaped (H) barrel-shaped joints after electromigration test. (F-H) Reproduced with permission from Ref. [128]. Copyright 2021, IEEE. (I) Effect of Cu pillar height on current crowding ratio. (J) Comparison of electromigration lifetimes between Cu pillar bump joints and flip-chip bump joints made of different materials. (I-J) Reproduced with permission from Ref. [129]. Copyright 2011, IEEE.

and 125°C . The results showed that the distribution characteristics of IMCs were highly dependent on grain orientation: when the c-axis of the Sn grain formed a large angle with the current direction, Cu atom migration was suppressed, IMC accumulation slowed, and joint stability improved; whereas a small angle resulted in significant Cu dissolution on the cathode side and extensive IMC accumulation (Fig. 12D and E).

In addition to the effects of barrier layers and bump composition, joint shape, solder volume, and copper pillar height also influence electromigration performance to varying degrees. For example, Fan et al. [128] investigated the influence of joint geometry on

electromigration behavior. Cylindrical joints exhibited more uniform current density and stress distribution, resulting in superior electromigration reliability and longer lifetimes (Fig. 12F). In contrast, hourglass-shaped joints had high current density at the neck region, leading to rapid IMC growth and crack formation, thus showing poorer reliability (Fig. 12G). For barrel-shaped joints, the current density and stress distribution also affected reliability; cracks mainly formed at the interface between the copper pillar and IMC, giving them shorter lifetimes than cylindrical joints (Fig. 12H). Park [132] investigated the effect of solder volume on electromigration behavior. In joints with

larger solder volumes, the Ni barrier layer was completely consumed because sufficient solder reacted with the Ni layer. In contrast, in structures with smaller solder volumes, the Ni layer was only partially consumed due to limited solder availability. As a result, SnAg_{2.5} micro-bumps with smaller solder volumes exhibited better electromigration reliability owing to the suppression of IMC formation. Yoo [133] compared the electromigration performance of two UBM structures (5 μ m Cu layer and 10 μ m Cu layer) and CPBs with a pillar height of 60 μ m under a current density of 5.09×10^4 A/cm². The lifetime distribution followed the order: CPB > Cu (10 μ m) > Cu (5 μ m). This indicates that CPB joints exhibit the best electromigration reliability, and also confirms that increasing the Cu layer thickness can effectively enhance the electromigration lifetime of bumps. Based on this, Syed [129] further performed finite element simulations to study the effect of Cu pillar height on the current crowding phenomenon (Fig. 12I). When the pillar height was 5 μ m, the current was mainly concentrated on the left side of the pillar, resulting in the highest current crowding ratio (the ratio of maximum to average current density). As the pillar height increased, the current distribution became more uniform and reached an optimal state at a height of 35 μ m. However, when the height further increased to 50 μ m, the crowding ratio slightly rose again, and the current density peak shifted toward the bump center. These results demonstrate that optimizing the Cu pillar height can mitigate current crowding and thereby improve electromigration resistance.

To evaluate the EM lifetime of chips, the commonly used predictive model is the Black's equation:

$$\text{MTTF} = AJ^{-n} \exp\left(\frac{E_a}{KT}\right) \quad (1)$$

where MTTF is the mean time to failure, A is a material- and geometry-dependent constant, J is the current density, n is the current density exponent, E_a is the activation energy, K is the Boltzmann constant, and T is the absolute temperature. By conducting experiments under various temperatures and current densities, the failure lifetime distribution can be obtained, and the parameters E_a and n can be fitted to predict the device lifetime under actual operating conditions. Under laboratory conditions, the electromigration lifetime of different interconnect structures can be evaluated by defining appropriate failure criteria.

It should be noted that Black's equation is an empirical model and does not explicitly account for the underlying physical mechanisms of electromigration. In recent years, physics-based models have been developed to provide a more comprehensive description of electromigration behavior by considering coupled effects such as atomic flux divergence, stress evolution, and multi-physics interactions. Representative work by X. J. Fan and co-workers has demonstrated improved predictive capability through such approaches [134–136]. Nevertheless, most studies on CPB interconnections still rely on experimental evaluation combined with empirical lifetime estimation. For example, Shao et al. [137] investigated the electromigration reliability of CPB joints under a current density of 1.5×10^4 A/cm² at 125 °C. Owing to the mitigation of current crowding in the UBM layer and the higher current-carrying capability of the Cu pillar, the joints exhibited a lifetime of 3387 h, which is significantly longer than that of conventional flip-chip bumps. In addition, Syed et al. [129] compared the electromigration behavior of conventional flip-chip solder joints (high-Pb, eutectic Sn-Pb and Sn-Ag) with CPB joints, and found that their resistance to electromigration improved sequentially in that order (Fig. 12J). Further tests on Cu pillar SnAg bumps with a diameter of 30 μ m revealed no failure even after more than 10,000 h of testing, demonstrating the excellent electromigration resistance of CPB joints [138].

In summary, electromigration failure is influenced by multiple interacting factors, suggesting that systematic enhancement of electromigration reliability in CPB joints can be achieved through optimizing geometric design and controlling crystal orientation. However, in practical applications, the packaging interconnect structure of chips is

simultaneously affected by electrical current and Joule heating, indicating that coupled multi-physics analyses are required to improve the accuracy of reliability prediction.

5. Summary and outlook

CPB technology has undergone more than a decade of continuous development and has become an important research focus in high-density interconnection due to its excellent electrical and thermal conductivity and adaptability to various interconnect pitches. Although material systems and fabrication processes have been continuously optimized, further improvements in performance and expansion of application scope are still required to meet the increasing demands of advanced packaging.

In terms of fabrication, the electroplating-reflow process remains the dominant industrial approach due to its maturity and scalability. However, the intrinsic limitations of solder materials and their melting behavior during reflow pose challenges for high-density interconnections. To address these issues, solid-state bonding strategies based on Cu and Ag micro-nano material systems have been proposed, offering low-temperature bonding capability and improved high-temperature stability. Nevertheless, these emerging approaches are still largely at the experimental stage and require further process optimization for practical applications.

From the perspective of interconnection processes, adapting to continuously shrinking pitches and enabling low-temperature bonding are key development directions. While reflow soldering is suitable for large-scale manufacturing, it faces increasing challenges such as bridging and warpage under fine-pitch conditions. Advanced techniques such as TCB and LAB provide improved controllability for fine-pitch interconnections, but also impose higher requirements on process precision and equipment capability. Looking toward future sub-10 μ m interconnections, further scaling of conventional solder-based approaches will become increasingly difficult. In contrast, TCB-based strategies remain among the most promising solutions, particularly when combined with advanced material designs such as Ag-coated Cu pillars. Meanwhile, Cu-Cu direct bonding is expected to play a key role in enabling ultra-fine-pitch interconnections, although it requires stringent surface preparation and bonding conditions. Future progress will rely on improved interface control, alignment accuracy, and process integration.

In terms of reliability, various testing methods, including HTST, TCT, TST, and EM tests, have been widely used to evaluate CPB performance. However, most studies focus on single-factor conditions, which deviate from real operating environments. Therefore, the development of multi-factor coupled testing approaches and multi-scale predictive models will be essential for more accurate reliability assessment. Overall, continued advancements in materials, processes, and reliability evaluation will further enhance the performance of CPB technology and expand its applications in advanced packaging.

CRedit authorship contribution statement

Gaohui Xu: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Lele Yang:** Validation, Methodology, Investigation, Data curation. **Su Ding:** Writing – review & editing, Methodology, Formal analysis. **Jiefei Gu:** Writing – review & editing, Methodology. **Lei Su:** Writing – review & editing, Methodology. **Wanli Li:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Ke Li:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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